Engine Model(s): PT6A

73-10-03

FUEL LINES - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00, STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, 0.025 inch diameter, and will not be called out in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC03-001	Oil, Engine Lubricating
PWC05-061	Cloth, Abrasive Coated
PWC05-101	Cloth, Abrasive
PWC11-027	Solvent, Petroleum
PWC11-031	Cleaner, Engine

3. Special Tools

Not Applicable

4. Fixtures, Equipment and Supplier Tools

Not Applicable

5. Removal/Installation

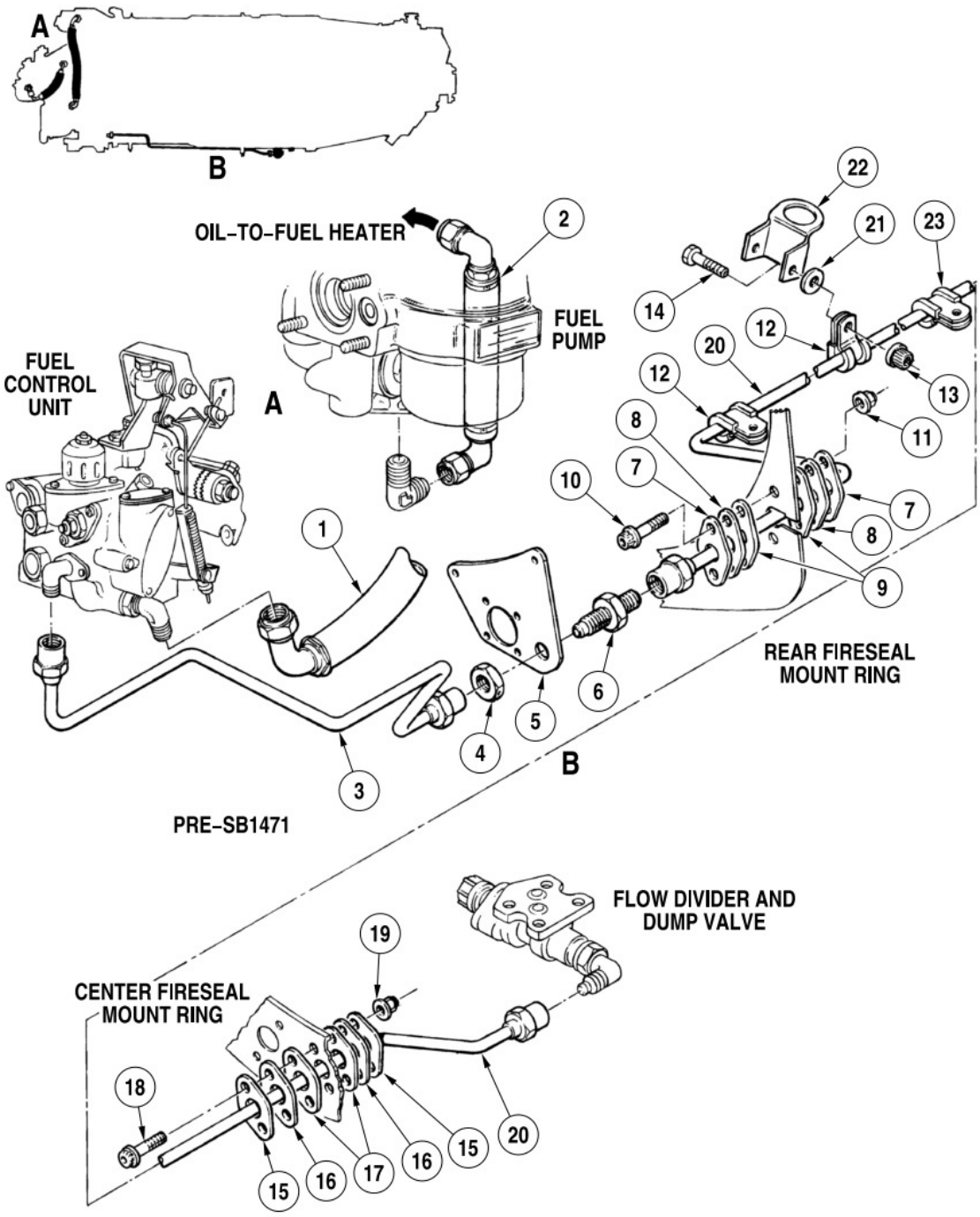
CAUTION: OBSERVE ALL FIRE AND SAFETY PRECAUTIONS.

A. Removal of Fuel Delivery Tubes (PT6A-21 Engines) (Ref. Fig. 201)

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Figure 201 Removal/Installation of Fuel Delivery Tubes

(SHEET 1 OF 2)

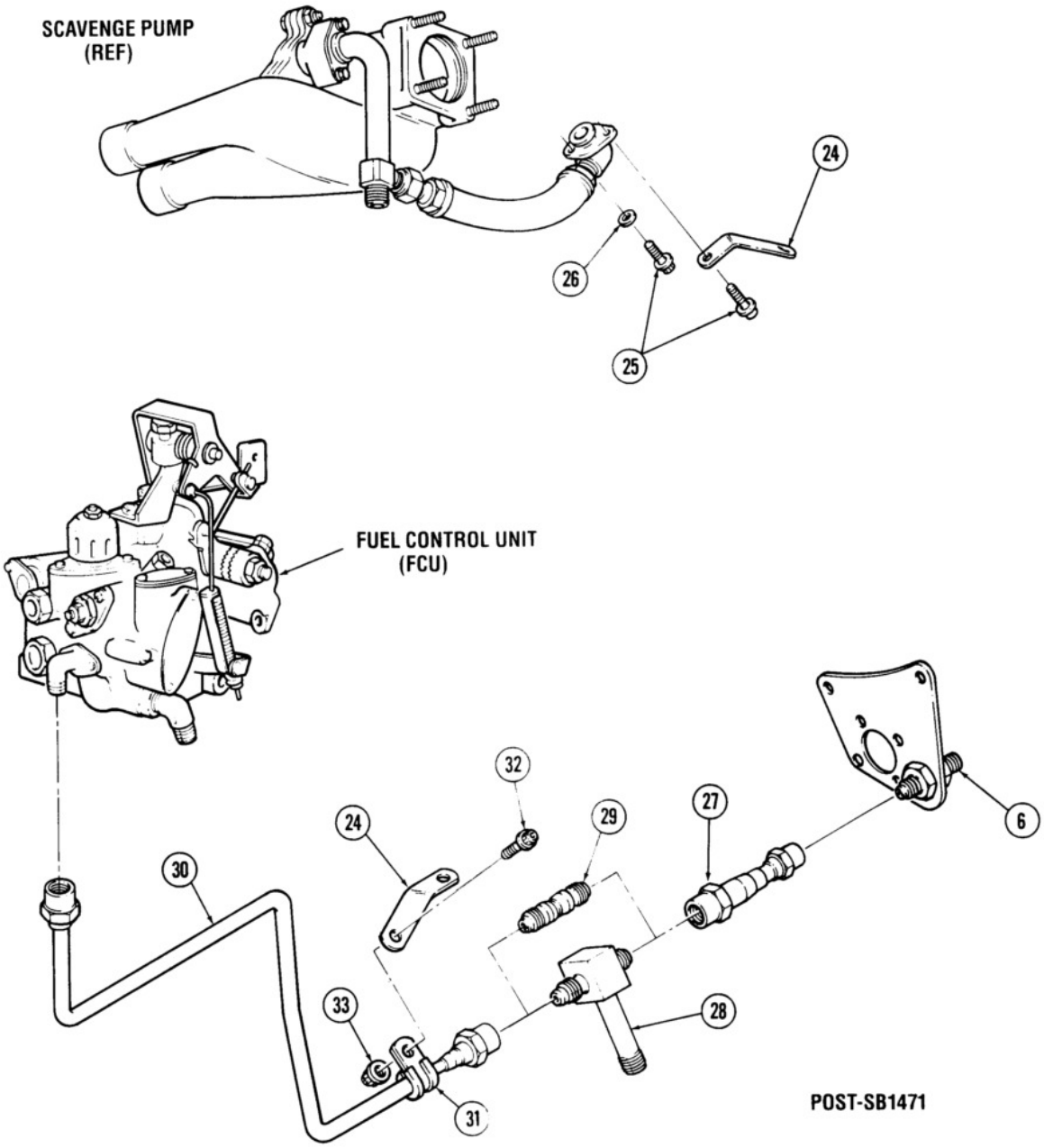


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PT6A-21

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1. Fuel Hose (fuel pump FCU)
2. Fuel Hose (fuel pump fuel heater)
3. Fuel Tube
4. Jam Nut
5. Bracket
6. Coupling
7. Retaining Plate
8. Insulation
9. Seal
10. Bolt
11. Self-locking Nut
12. Loop Clamp
13. Self-locking Nut
14. Bolt
15. Retaining Plate
16. Insulation
17. Seal
18. Bolt
19. Self-locking Nut
20. Fuel Tube
21. Washer(s)(two max)
22. Support Bracket
23. Loop Clamp (glow plug installation only)
24. Bracket
25. Bolt
26. Washer
27. Fuel Tube Assembly
28. Fuel Flow Transmitter
29. Fuel Pressure Tube Assembly
30. Primary Fuel Tube Assembly
31. Clamp
32. Bolt
33. Nut

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(1) Disconnect coupling nuts at each end of flexible fuel hose (2) from oil-to-fuel heater and fuel pump. Remove hose and cap all openings.

(2) Disconnect coupling nuts at each end of flexible fuel hose (1) from fuel pump and fuel control unit (FCU). Remove hose and cap all openings.

(3) Disconnect coupling nuts at each end of fuel tube (3) from FCU and bulkhead coupling (6) at fuel pressure tube bracket (5). Remove tube and cap all openings.

(4) Remove self-locking nuts (11) from bolts (10) at rear fireseal mount ring, and remove fuel line seals (9) and seal retaining plate insulation (8) from tube.

(5) Remove self-locking nuts (19) from bolts (18) at center fireseal mount ring, and remove fuel line seals (17) and seal retaining plate insulation (16) from tube.

(6) Disconnect coupling nuts at each end of fuel tube (20) from bulkhead coupling (6) at fuel pressure tube bracket (5) and elbow fitting on flow divider and dump or purge valve.

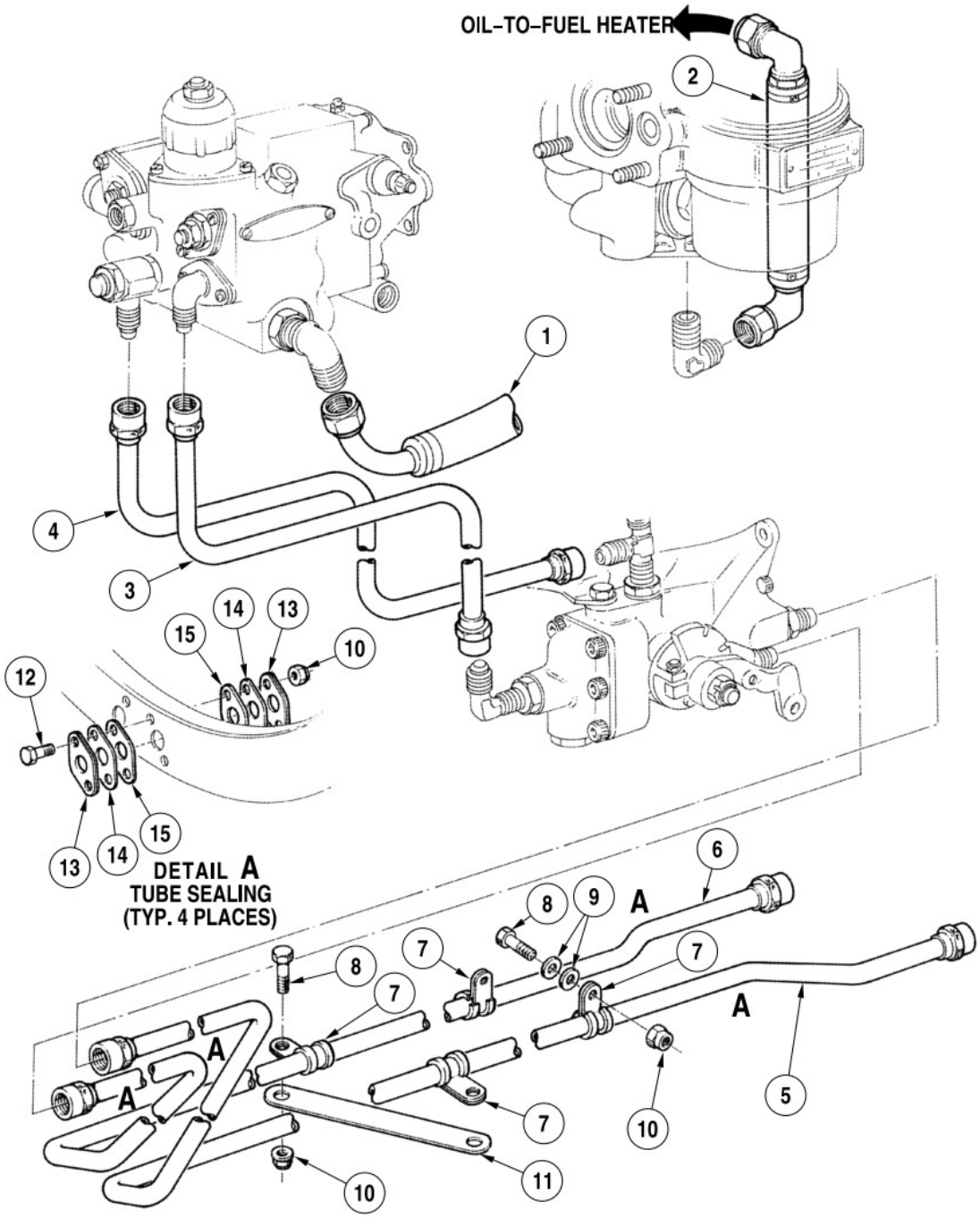
(7) Remove self-locking nut (13) and bolt (14) that secure loop clamp (12) and washer (21), if fitted, to support bracket (22) at rear fuel drain valve. Also, remove nuts and bolts securing fuel line loop clamps (12), and (23) if fitted, to similar loop clamps on ignition cable (Ref. 74-20-01). Remove fuel tube (20).

NOTE: Unless fuel tube is to be replaced, loop clamps (12) andseal retaining plates (7) and (15) may remain installed.

B. Removal of Fuel Delivery Tubes (PT6A-27 and PT6A-28 Engines) (Ref. Fig. 202)

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Figure 202 Removal/Installation of Fuel Delivery Tubes



PT6A-27 and PT6A-28

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1. Fuel Inlet Hose
2. Fuel Hose
3. Fuel Pressure Rear Tube
4. Fuel Bypass Tube
5. Fuel Pressure Primary Tube
6. Fuel Pressure Secondary Tube
7. Loop Clamp
8. Bolt
9. Washer
10. Self-locking Nut
11. Back Plate
12. Bolt
13. Retaining Plate
14. Insulation
15. Seal

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(1) Disconnect coupling nuts at each end of flexible fuel hose (2) from oil-to-fuel heater and fuel pump. Remove hose and cap all openings.

(2) Disconnect coupling nuts at each end of flexible fuel hose (1) from fuel pump and fuel control unit (FCU). Remove hose and cap all openings.

(3) Disconnect coupling nuts at each end of fuel tubes (3) and (4) from FCU and starting flow control. Remove tubes and cap all openings.

(4) Remove self-locking nuts (10) and bolts (12) at center and rear fireseal mount rings, and remove fuel line seals (15) and seal retaining plate insulation (14) from tube.

(5) Remove self-locking nuts (10), bolts (8) and washers (9) from loop clamps (7).

(6) Disconnect coupling nuts at each end of fuel tubes (5) and (6) from starting flow control and fuel inlet adapter. Remove tubes and cap all openings.

NOTE: Unless fuel tube(s) is to be replaced, loop clamps (7) andseal retaining plates (13) may remain installed.

C. Installation of Fuel Delivery Tubes (PT6A-21 Engines) (Ref. Fig. 201)

(1) If replacement rear fuel tube (20) is being fitted, install four seal retaining plates (7) and (15) on tube; install two loop clamps (12), and (23) if fitted, on tube, clamps to be located between two center retaining plates.

(2) Locate fuel tube (20) between bulkhead adapter and fuel inlet adapter, and secure coupling nuts. Do not torque nuts at this stage.

(3) Secure front loop clamp (12) to support bracket (22) with bolt (14) and self-locking nut (13). Tighten nut and torque to 36 to 40 lb.in.

NOTE: Use washer(s) (21) between clamp and support bracket toprevent any misalignment. No side loading on tube is permissible.

(4) Secure rear loop clamp (12) to corresponding loop clamp on ignition cable with bolt and self-locking nut. Tighten nut and torque to 36 to 40 lb.in.

(5) Install fuel line seals (9) and seal retaining plate insulations (8) on each side of rear fireseal mount ring and secure with seal retaining plates (7), nuts (11) and bolts (10). Tighten nuts and torque to firm contact plus 180 degrees.

NOTE: Bolt heads must be installed on air inlet side of mountring.

(6) Similarly, install fuel line seals (17) and seal retaining plate insulations (16) on each side of center fireseal mount ring. Secure with seal retaining plates (15), nuts (19) and bolts (18). Tighten nuts until firm contact is made plus 180 degrees.

(7) Install fuel tube (3) between elbow fitting on FCU and bulkhead adapter (6), and secure with coupling nuts. Torque nuts at each end of fuel tubes (3 and 20), 90 to 100 lb.in., and safety wire.

(8) For Post-SB1471 Engines:

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CAUTION: MAKE SURE THAT ALL BOLTS AND FUEL TUBE COUPLING NUTS ARE LOCKWIRED.

(a) Install bracket (24) in aligned position on rear mounting hole of scavenge oil hose flange and secure with bolt (25) fingertight. Secure forward flange mount hole with washer (26) and bolt (25). Torque forward bolt 24 to 36 lb.in.

(b) Locate fuel tube assembly (27) on coupling (6). Tighten nut and torque 90 to 100 lb.in.

(c) Install fuel flow transmitter (28) to fuel tube (27). Tighten nut fingertight.

NOTE: If fuel flow transmitter is unavailable, install fuelpressure tube assembly (29) in place of the transmitter, following the same instructions.

(d) Locate primary fuel tube assembly (30) to fittings on fuel flow transmitter (28) and fuel control unit (FCU). Tighten coupling nut of flow transmitter fingertight. Torque coupling nut at FCU 90 to 100 lb.in.

(e) Install clamp (31) on fuel tube (30) near flow transmitter. Align holes in clamp (31) and bracket (24), install bolt (32) and secure with nut (33). Torque nut 36 to 40 lb.in. Torque bolt (25) in rear mounting hole of scavenge oil hose flange 24 to 40 lb.in.

(f) Ensure correct clearance angle maintained. Torque coupling nuts securing fuel flow transmitter 450 to 500 lb.in. and safety wire.

CAUTION: IT MAY BE NECESSARY TO CHANGE ANGULAR POSITION OF FUEL CONTROL INLET ELBOW TO ACHIEVE CORRECT ALIGNMENT OF HOSE (1), IN WHICH EVENT INSTRUCTIONS IN STANDARD PRACTICES (REF. 70-00-00) MUST BE OBSERVED TO AVOID DAMAGE TO PREFORMED PACKING, WITH RESULTANT LEAKAGE.

(9) Install fuel hose (1) between FCU and fuel pump, and secure coupling nuts. Torque nuts 170 to 200 lb.in., and safety wire.

(10) Install fuel hose (2) between fuel pump and oil-to-fuel heater, and secure coupling nuts. Torque nuts 450 to 500 lb.in., and safety wire.

D. Installation of Fuel Delivery Tubes (PT6A-27 and PT6A-28 Engines) (Ref. Fig. 202)

(1) If replacement of either/or both fuel tubes (5 and 6) is being carried out, install seal retaining plates (13) and loop clamps (7) on tube(s).

(2) Locate fuel lines (5) and (6), through center and rear fireseal mount rings, and secure coupling nuts to respective fittings onfuel inlet adapter and starting flow control. Do not torque coupling nuts at this stage.

(3) Install loop clamps (7) on fuel tubes (5) and (6) to respective arms of support bracket at rear fuel drain valve and install bolts (8) and self-locking nuts (10).

NOTE: Washers (9) (two max. each location) must be installed, as required, to prevent tube distortion due to side loading.

(4) Torque coupling nuts 90 to 100 lb.in., and safety wire. Torque loop clamp self-locking nuts 36 to 40 lb.in.

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(5) Install fuel line seals (15) and seal retaining plate insulation (14) on fuel tubes (5) and (6). Secure seals, insulation and plates (13) to respective fireseal mount rings with bolts (8) and self-locking nuts (10). Torque nuts until firm contact is made plus 180 degrees.

NOTE: Bolt heads must be installed on air inlet side of mount rings.

CAUTION: WHEN INSTALLING FUEL TUBE (3) ENSURE THAT SHORTER END ISCONNECTED TO FCU. INCORRECT INSTALLATION MAY RESULT IN CHAFING BETWEEN TUBE AND ADJACENT AIRFRAME EQUIPMENT.

(6) Install fuel tubes (3) and (4) between respective fittings on FCU and starting flow control. Tighten coupling nuts, torque to 90 to 100 lb.in., and lockwire.

CAUTION: IT MAY BE NECESSARY TO CHANGE ANGULAR POSITION OF FUELCONTROL INLET ELBOW TO ACHIEVE CORRECT ALIGNMENT OF HOSE (1). IN THIS EVENT, OBSERVE INSTRUCTIONS IN STANDARD PRACTICES (REF. 70-00-00) TO AVOID DAMAGE TO PREFORMED PACKING WITH POSSIBLE RESULTANT LEAKAGE.

(7) Install fuel hose (1) between FCU and fuel pump, and secure coupling nuts. Tighten nuts, torque to 270 to 300 lb.in., and lockwire.

(8) Install fuel hose (2) between fuel pump and oil-to-fuel heater and secure coupling nuts. Tighten nuts, torque to 450 to 500 lb.in., and lockwire.

6. Cleaning/Painting

A. Clean Flexible Hoses and Fuel Line Tubing

(1) Clean the exterior of tubing by spraying with petroleum solvent (PWC11-027) or (PWC11-031) and brushing with a soft bristle brush, as necessary. Dry using compressed air, observing all normal ventilation and fire safety precautions.

(2) Clean tubing interior by pressure flushing with petroleum solvent. Dry using clean, filtered compressed air, taking particular care that no foreign matter remains in tubing.

(3) Cap the tube ends immediately after cleaning and until the tube is to be installed on the engine to prevent ingress of foreign matter.

7. Inspection/Check

A. Flexible Fuel Hoses

(1) Examine the hose for cuts, crimping or buckling. Hoses with such defects must be replaced.

(2) Examine the coupling nuts for cracks. Replace the hoses as necessary.

B. Fuel Lines

(1) Inspect tubing:

(a) Scratches. Minor scratches having no appreciable depth are acceptable. Scratches to a depth of 0.005 inch must be blended out. Replace tubing with scratches over 0.005 inch deep.

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(b) Nicks. Individual nicks up to 0.062 inch long by 0.010 inch wide and 0.003 inch deep can be ignored. Nicks to a maximum depth of 0.005 inch must be blended out to remove sharp edges. Replace tubing with nicks greater than 0.005 inch deep.

(c) Dents. Round bottom dents on straight sections of tubing are permitted provided the length and depth are not greater than 10 percent of the normal outside diameter of the tube. No more than one dent to a maximum depth per 12 inch length of the tube is acceptable. Dents on tube bends, which cause restriction by flattening and local weakening, are not acceptable.

(d) Pitting. Minor isolated pitting up to 0.003 inch deep is acceptable. Clusters of pitting should be blended out to a maximum depth of 0.005 inch. Tubing must be replaced if pitting exceeds 0.005 inch deep.

(e) Corrosion. Staining on tubing and surface corrosion is acceptable if removable by light polishing with a crocus cloth and oil.

8. Approved Repairs

A. Repair of Fuel Tubes

(1) Smooth out small nicks with abrasive cloth (PWC05-061).

(2) Remove stains and/or corrosion by polishing with abrasive cloth (PWC05-101) and oil (PWC03-001).

9. Adjustment/Test

A. Procedure

(1) Start engine, and do checks associated with Installation/Replacement of Fuel Lines (Ref. 71-00-00, ADJUSTMENT/TEST).

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